

Pattern-of-life Analysis and Deviations for Bluetooth and Bluetooth Low Energy Devices Spring 2021

The University of Texas at Dallas
Team All Your Cookie Are Belong to US

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Project Summary

Description

Many commercial products exist to monitor traditional 802.11 wireless networks, but there has been limited research aimed at monitoring other wireless protocols such as Bluetooth and Bluetooth Low Energy (BLE). Many devices have built-in Bluetooth communications, to include cell phones, smartwatches, fitness devices, lighting systems, refrigerators, and countless others. As Bluetooth devices begin to take over a large portion of the network communications in our homes and workplaces, how do we monitor these devices to ensure they're behaving as they should?

Motivation

Due to the rise of IOT devices, bluetooth networks are now part of a significant portion of all communication networks. This potentially creates a new security hazard of devices utilizing bluetooth as their primary form of attack. One approach to this problem is to create a lifetime model of such devices to determine whether they are acting in a normal or abnormal fashion. We are interested in developing a detection system to analyze the pattern of life of a bluetooth device and improve the ability to monitor such attacks by detecting them before or as they occur.

Organization

Deliverables

- Proposals and Literature Review
- Progress Reports and Representation
- Final Report and Presentation
- Documentation and Code delivery

Schedule/Milestones

2/12: Project Proposal and Technical Approach
2/19: Whitepaper research and pattern of life feature identification
3/05: Collect data and extract pattern of life features
3/25 (MIDTERM): Develop and implement pattern of life profile
4/09: Produce anomalous traffic and test against pattern of life
4/23: Implement pattern of life for multiple (>3) devices
5/06: Begin machine learning for automated pattern of life development

Budget

For this project, we expect an initial low cost of purchasing several bluetooth devices and accompanying bluetooth networking hardware. We do not expect the total cost to exceed \$1000

Team Approach and Justification

Technical Approach

The goal of the project is to be able to detect anomalous behavior of bluetooth devices through analysis of their emitted signals. By building a pattern of life model for a specific device, we should be able to determine if certain behaviors of said device is expected and intended behavior. In the case that behavior is atypical of such a device, such anomalies are potential indicators of malicious behavior. In our model, the device would then be flagged and an alert raised.

To accomplish our goal, we plan to use an ubertooth device to collect bluetooth packets from various types of bluetooth devices. The devices we have selected are a smartwatch (Galaxy Watch 3), a fitness tracker (Fitbit), wireless earphones (Galaxy Buds Pro), and various types and models of smartphones. We plan on using a machine learning based approach to analyze the signals we collect and build an appropriate model to predict whether a device is acting in accordance to its normal behavior.

Once we have built such a model, we hope to use the model to analyze devices in real time as they are transmitting data. We plan to test our model by simulating the effects of a malicious device attacking some system that the model is aiming to protect. To this end, we will use a custom application running on the test devices to send malicious bluetooth packets. We will then determine the effectiveness of our model by how well it is able to identify and isolate the malicious attacking device.

Midterm Report

We have successfully set up a virtual machine and captured several instances of bluetooth traffic. The traffic captured includes advertising packets from the bluetooth advertising channels 37, 38, and 39. Additionally, we have gathered several instances of captures from data channels and are examining the gathered data to see if we are able to pair the information from the already gathered advertising packets to the data gathered from the data channels.

```

systemtime=1614586906 freq=2426 addr=8e89bed6 delta_t=419.745 ms
c3 0c ad ab f9 44 f8 42 78 5f f6 04 34 65 5d 8d 99
Advertising / AA 8e89bed6 (valid)/ 12 bytes
  Channel Index: 38
  Type: SCAN_REQ
  ScanA: 42:f8:44:f9:ab:ad (random)
  AdvA: 65:34:04:f6:5f:78 (random)

  Data: ad ab f9 44 f8 42 78 5f f6 04 34 65
  CRC: 5d 8d 99

systemtime=1614586906 freq=2426 addr=8e89bed6 delta_t=272.494 ms
c3 0c ad ab f9 44 f8 42 78 5f f6 04 34 65 5d 8d 99
Advertising / AA 8e89bed6 (valid)/ 12 bytes
  Channel Index: 38
  Type: SCAN_REQ
  ScanA: 42:f8:44:f9:ab:ad (random)
  AdvA: 65:34:04:f6:5f:78 (random)

  Data: ad ab f9 44 f8 42 78 5f f6 04 34 65
  CRC: 5d 8d 99

```

To examine the data we have gathered so far, we have created several scripts to break down the data using the scapy libraries. Upon running the script, we have noticed that several of the captured packets contain information that appears to identify the manufacturer of the device.

Next Plan

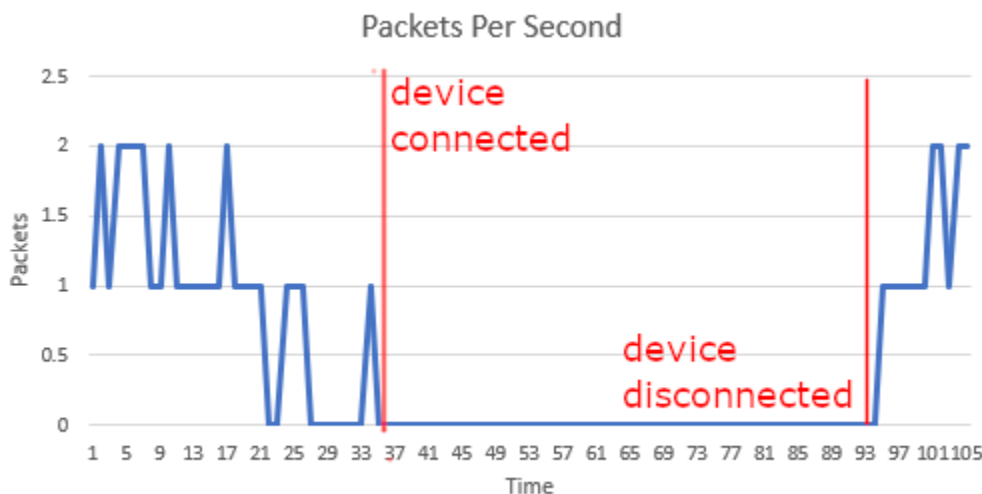
We plan to continue to gather additional captures of bluetooth packets in order to increase the overall amount of data. Once the amount of data we have collected is sufficient, we plan to cease data collection and move onto the analysis phase of the project. We plan to examine several features of the captured bluetooth data in order to create a pattern of life for the devices we are working with such as packet density, packet frequency, and packet type. Additionally, we plan to look into identifying packets based on manufacturer specific information included in the fields of the bluetooth packets.

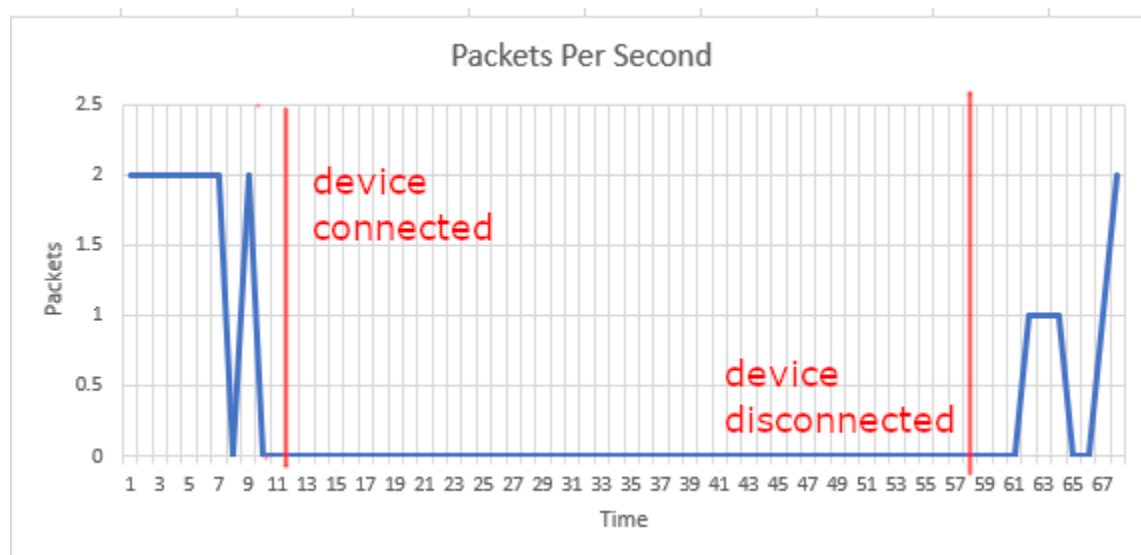
Final Report

Final Outcome

Using manufacturer specific information included in the bluetooth packet captures (specifically a unique one byte value found in the advertising data), we were able to identify the devices which produced the packets. We used this information obtained from the bluetooth captures to perform an analysis upon the captured data to determine a pattern of life. By examining the rate of communication on the advertising channels, we were able to see that once a device has paired with another device and started to communicate within the data channels, the devices would no longer send advertising packets. Thus, by looking for large gaps in an otherwise consistent ping of advertising packets, we were able to determine which devices were connected and which were not.

```
vishalmdipadga@Vishals-MacBook-Pro.local:/Users/vishalmdipadga/CS6301 $ python detMan.py Channel37Disconnect.pcap
Number of Microsoft Packets: 0
Number of Apple Packets: 0
Number of Samsung Packets: 404
vishalmdipadga@Vishals-MacBook-Pro.local:/Users/vishalmdipadga/CS6301 $ python detMan.py Channel38Disconnect.pcap
Number of Microsoft Packets: 0
Number of Apple Packets: 0
Number of Samsung Packets: 585
vishalmdipadga@Vishals-MacBook-Pro.local:/Users/vishalmdipadga/CS6301 $ python detMan.py Channel39Disconnect.pcap
Number of Microsoft Packets: 0
Number of Apple Packets: 0
Number of Samsung Packets: 688
```





Lessons Learned

Bluetooth communications occur in two different channels. When devices are not initially paired, they communicate on one of three advertising channels, Channel 37, Channel 38, and Channel 39. Once a connection is established within the advertising channel, the paired devices then move to one of many data channels and resume communication as such. Thus, to peek in on bluetooth traffic, one must first survey the advertising channels to determine what type of devices are open for connection. From the information gained there, a potential attacker can then tune into the appropriate data channel to eavesdrop further.

Pattern-Of-Life Database

Using the bluetooth packet captures we gathered, we created a database to store the packet information for a specific bluetooth device. Each database has two tables, an Advertising table and a Data table. The Data table was meant to be based on the GATT profile found in ATT packets that a bluetooth device transmits (on data channels) to advertise services it has available. Each column in the data table would correspond to a service and/or characteristics associated with the service that the device provided. However, we were unable to capture ATT packets. The Advertising table stores statistics about advertising packets for a specific bluetooth device. Specifically, each row in the database corresponds to one hour of packet capture (this time interval can be adjusted if needed, for example, looking at thirty minutes of packet capture). The information we gathered was the start and end time of the packet capture, average time between advertising packets, the average packet length, the percentage of each type of advertising packet, and the number of MAC addresses used by the device.

The goal of the database was to try to establish a profile of what benign bluetooth advertising activity looks like. Once this bluetooth data had been gathered over a long period of time, our goal was to use machine learning to see if we could identify anomalous advertising behavior based on the benign advertising behavior we had previously observed and stored in our database. The machine learning algorithm we were looking into using was local outlier factor (LOF) which has been used in some studies on malware detection. The LOF would be trained on purely benign data. When the time comes

to classify a data point, it would compare the density of that point (which is based on distance between neighboring points) to the density of the points that were used to train the LOF. A data point with a low density would be treated as anomalous. In our case, the rows in our database would be treated as benign data points that would be used to train the LOF. Ideally, malicious bluetooth devices would exhibit advertising behavior that is different from the advertising behavior stored in the database. We did not manage to implement/train an LOF, however, this could be considered for future work.

deviceid	capture_starttime	capture_endtime	advertising_packet_frequency	average_packet_length	percentage_adv_ind	percentage_adv_nonconn_ind	num_mac_addresses
117	1619117401617	1619121097637	0.744055944056	43	100	0	1
117	1619121097637	1619124963104	0.75326058025	43	100	0	1
117	1619124963104	1619128633133	0.761081081081	43	100	0	1
117	1619128633133	1619132411374	0.754330879168	43	100	0	1
117	1619123876961	1619127669802	0.732584269663	43.0058400719	99.9550763702	0.0449236298293	2
117	1619127669802	1619131517041	0.733686067019	43.0028653295	99.9981367221	0.00186327788591	1
117	1619131517041	1619135117084	0.7482898227	43.0018146287	99.999931673	6.83270218522e-05	1
117	1619135117084	1619138854801	0.748677534276	43.0014032815	99.9999968942	3.10577372056e-06	1
117	1619138854801	1619142683889	0.7460776218	43.0010734043	99.9999998946	1.05387638974e-07	1
117	1619142683889	1619146549371	0.748938360571	43.0008363894	99.999999997	2.98379498793e-09	1
117	1619146549371	1619150226450	0.750954198473	43.0006891068	99.9999999999	8.71944765614e-11	1
117	1619150226450	1619153850822	0.752063323519	43.0006274131	100	4.46007552744e-12	1
117	161915488413	1619119131430	0.795530726257	43.1042442293	98.5107967238	1.48920327625	2
117	1619137293853	1619140959443	0.753039513678	43	100	0	1
117	1619140959443	1619144869790	0.782335025381	43	100	0	1
117	1619144869790	1619148488466	0.797357482185	43	100	0	1
117	1619148488466	1619152317079	0.777810885975	43	100	0	1

Future Work

The mainline for future work would be using the concept of provenance as a way to handle threat detection instead of the traditional single based anomaly approach which is prone to false alarms. The idea of provenance is to consider the forward and backward context of a specific event. With the forward and backward context graphs captured and merged together, we can then extract paths by choosing the k number of rarest paths which is determined by a database summarizing a device's previous activity also known as a frequency database. Once we have the paths we can then convert them into a vector space using doc2vec and determine the maliciousness of the execution of a possible device and report it potentially. This requires building an event and frequency database which takes a little too much time to capture events for just one particular device so we leave this problem up for potential future work.

Literature Review

1 Fingerprinting Bluetooth-Low-Energy Devices Based on the Generic Attribute Profile

In this paper, the authors discuss how Bluetooth Low Energy (BLE) devices can be fingerprinted using the Generic Attribute Profile (GATT) of a BLE device. According to the paper, GATT is a data structure that the device advertises using the Attribute (ATT) protocol. This advertised profile can be used to gain sensitive information about a user. BLE devices can be Peripheral (devices that can accept or reject incoming connections) or Central (devices that look for Peripheral devices to connect to). Peripheral devices broadcast advertisement packets to allow Central devices to discover them. BLE devices have 4 types of addresses: Public, Random Static, Random Non-resolvable, and Random Resolvable. The Public address is the device's physical MAC address while the other 3 are randomly generated. A GATT profile is made up of attributes that can be classified as services or characteristics, each with their own UUID. Characteristics consist of an attribute handle (position of a characteristic in a profile), a value (value of the characteristic), and properties (ATT operations that can be performed). Services are composed of two handles that let us know the characteristics of a service. The study presented in the paper involved analyzing the GATT profile from 13925 BLE devices. A Raspberry Pi with 4 USBs was used (1 to scan for Peripherals and the other 3 to connect to the discovered Peripherals and retrieve the GATT profile). The authors then attempted to create a fingerprint of the device based on the GATT profile. The authors discovered that GATT profiles can be composed of a large number of services and characteristics. Furthermore, some characteristics and their value is dependent on the device's function. To create a fingerprint, the authors made use of a service's UUID and handles as well as a characteristics UUID, handle, properties, and value. They then calculated the entropy of each of these artifacts. The study calculated 3 values for 8 common services and characteristics. Entropy (identifying bits of the artifact), Stability (proportion of devices that were observed several times while their artifact value remained constant), and Affected devices (portion of devices that reported given artifact). They found that the stability of their proposed fingerprint was stable, depending on the type of address used by a device some characteristics provide less bits of information, and characteristics provide more bits of information when compared to services. The authors also analyzed anonymity sets (set of devices with the same fingerprint). Bigger anonymity sets meant more privacy. They found that based on the type of address used by the device, the anonymity set size varied.

2 A Comprehensive Study of Bluetooth Fingerprinting-based Algorithms For Localization

The paper analyzes a variety of methods for fingerprinting and locating bluetooth devices. The paper focuses on three algorithms, kNN, neural networks, and SVM. The paper notes that kNN is a good algorithm for real life application use in locating the position of a device and fingerprinting it.

Fingerprinting as discussed is the ability to estimate the location of a device by matching it to the previous known locations of bluetooth signals. In the paper, a mobile phone was used to collect bluetooth signals and five other bluetooth devices were placed at varying distances from the collection point. The devices tested were a bluetooth adapter, an iPad, an iPhone, a Vodafone 845 Android phone, and a Macbook Air. Experiments were performed with the three algorithms in an attempt to determine the position of the test devices.

In terms of model training time, SVM required the most time to produce results. Neural networks took the second longest while kNN took the least amount of time while maintaining the highest accuracy at least when using kNN-r. The paper concludes that the kNN algorithm is the best algorithm among those tested with regards to fingerprinting bluetooth devices with the lowest error and shortest training time.

3 An investigation of Bluetooth security vulnerabilities

Bluetooth technology has improved as it evolved through the years. They are convenient and easy to use in our day to day lives. Like any other technology there are security flaws which means these devices are vulnerable to attacks. This paper talks about the vulnerabilities and the tools and techniques that are available to attackers to exploit these vulnerabilities. In this paper it also discusses techniques to mitigate risks and to protect data from these bluetooth devices.

A breakdown of this paper, there's a background overview about bluetooth in general. They analyze the security of Bluetooth and discuss the vulnerabilities. In this paper they also show their performance analysis and results of their experiment. They performed a hands-on investigation between two devices of their security vulnerabilities associated with the Bluetooth Low Energy Protocol. In this investigation, they installed a Ubertooth One tool, an open source hardware and software package. This device is used for monitoring Bluetooth traffic. Mitigation techniques were also discussed in this paper as well as its issues. Since bluetooth uses wireless communication, the signal transmission is vulnerable to being captured by an attacker.

Bibliography

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[3] P. Cope, J. Campbell and T. Hayajneh, "An investigation of Bluetooth security vulnerabilities," *2017 IEEE 7th Annual Computing and Communication Workshop and Conference (CCWC)*, Las Vegas, NV, 2017, pp. 1-7, doi: 10.1109/CCWC.2017.7868416.

Biographical Sketches of Investigators

Alex Armstrong

Alex Armstrong is a graduate level Computer Science major with a focus in Information Assurance from the University of Texas at Dallas. He is also a recipient of CyberCorps: Scholarship for Service. He has experience in system and network programming in regards to solving Cybersecurity issues. In particular, he has created an antivirus man-in-the-middle web proxy server to filter out malicious content from both secure (HTTPS) and non-secure (HTTP) webpages using a combination of C and Python. He has completed coursework in operating systems, computer networks, network security, computer architecture, and database security.

Seth Bedford

Seth Bedford is a graduate Computer Science student with a focus on Information Assurance at the University of Texas at Dallas. He has experience working on authn/authz systems at the National Security Agency (DoD), developing Python scripts for government use at the Cybersecurity and Infrastructure Security Agency (DHS), using C/C++ and Java through school projects, and developing with various web technologies through a mix of personal and academic projects. He is also the primary developer of ConnArtist, a simple web-based network traffic visualization tool that parses and displays network data using conntrack. He has also completed coursework in operating systems, computer networks, computer architecture, machine learning, and system security and malicious code analysis. Current coursework includes network security, information security, and data and applications security.

Troy Frazier

Troy Randall Frazier is a graduate computer scientist focusing within Digital Forensics and Cryptography. Troy has experience in many programming languages spanning all paradigms and can easily pick up more (Focuses on C/C++/Python). Internships focused on digital forensics with occasional cryptography (Usually for integrity in the forensic process). Troy has experience in all 7 layers of the OSI model (Wireshark for IP layer, TLS for transport layer, etc) as well as Ghidra and Ida for reverse engineering. Troy also is experienced at looking at assembly and machine code. Troy's extracurriculars focus on cryptography. Troy was a project lead for TexSaw2020 Crypto branch, UTD Crypto Mentor for CSG, and researches crypto systems as a hobby. Other experiences involve front-end/back-end programming, databases, network design, steganography, programming language design, and high-level research.

Vishal Madipadga

Vishal Madipadga is a graduate Computer Science student at the University of Texas at Dallas. He is a recipient of the CyberCorps Scholarship for Service. He has experience programming in C/C++, Java, and Python through school projects and when he worked as a Software Engineer Intern at L3Harris Technologies. He has experience analyzing Wireshark data through his experience pentesting and discovering a vulnerability with an IoT device. He has experience in reverse engineering (specifically reverse engineering C and C++ binaries) through participation in Capture The Flag (CTF) competitions which has given him exposure to tools such as Ghidra and gdb. He also has experience with databases through his work in the UT Dallas System Security Lab in the area of malware detection. He has completed coursework in System Security and Malicious Code Analysis, Network Security, Computer Networks, and Machine Learning.

James Wei

James Wei is a graduate student studying cyber security at The University of Texas at Dallas. He is a recipient of the CyberCorps Scholarship for Service (SFS) award and the Academic Excellence Scholarship (AES). He participated in a Research Experiences for Undergraduates (REU) program during his undergraduate studies. Specifically, the program focused on software safety and reliability with a heavy base in research and application. During the program, he lead a team to develop a secure online web application with the goal of helping a local charity store their data online.

He chose to study cyber security due to personal interest in the subject and in the first semester of his graduate studies he worked in a coop internship program with L3Harris Technologies. During his internship, he worked in developing a secure system for transferring messages between hardware processors in a radar system.

Evelyn Wong

Evelyn Wong is a Graduate Computer Science major, with a specialization in information assurance. She is currently a graduate student and working on a master's degree in computer science. She is also a recipient of the CyberCorps: Scholarship for Service. She has experience practicing and studying attacking techniques and vulnerabilities in computer and network systems. She has completed related coursework in software engineering, machine learning, computer architecture, computer network, and database design. She is currently taking web programming languages, network security, and Design and Analysis of Computer Algorithms.